

Framed IP Architecture & Walled Garden Redirection

Technical Guide: NAS vs. RADIUS IP Allocation & Subscription Lifecycle Workflows

1. The Core Concept: What is a Framed IP?

In broadband access networks (PPPoE & Hotspot) conforming to RFC 2865, **Framed-IP-Address** represents the IPv4 address assigned to the subscriber's terminal equipment (e.g. ONT, CPE, or home Wi-Fi router) for the duration of their point-to-point framed network session.

The Fundamental Question: *Should the IP address be allocated by the NAS (MikroTik) or by RADIUS (FreeRADIUS)?*

Industry Standard & Airlink Verdict:
For 95%+ of ISP deployments, the NAS (MikroTik) should always allocate dynamic subscriber IPs from its local pool. RADIUS should control *policy* (speed, validity, pool name, address list), but leave local IP leasing to the router hardware.

2. Architectural Comparison: NAS vs. RADIUS Allocation

Evaluation Criteria	NAS Allocation (MikroTik Pool)	RADIUS Allocation (sqlippool / DB)
Reliability & Leases	Zero Ghost Leases: MikroTik tracks physical sessions in kernel memory. If a line drops, the IP is freed immediately.	Ghost Lease Risk: If FreeRADIUS misses an Accounting-Stop packet (power cut, dropped UDP), the IP stays locked in DB forever.
Database Load	Zero DB Overhead: Authentication is a fast read-only query. IP allocation happens entirely on router hardware.	Heavy DB Locking: Every single dial-in must execute SQL transaction locks (`SELECT FOR UPDATE`) on the `radippool` table.
Multi-Tower Routing	Automatic Routing: Tower A uses `10.100.0.0/20`, Tower B uses `10.100.16.0/20`. Subnets are aggregated naturally.	Complex Dynamic Routing: RADIUS can lease an IP from a central block to any router, requiring host-route (/32) OSPF/BGP distribution.
Static Public IPs	Hybrid Override: Set `Framed-IP-Address` in RADIUS only for VIP accounts; MikroTik overrides its local pool automatically.	Centralized Management: All public IP blocks reside in the central database.

3. Analysis of the Walled Garden Subscription Redirection Workflow

The architecture diagram you provided represents the **ISP Walled Garden Pattern**. Instead of abruptly dropping the PPPoE connection with an `Access-Reject`, the subscriber is granted a **Restricted Session**.

State	RADIUS Decision	MikroTik Pool / Subnet	Network Access & Customer Experience
ACTIVE	Returns standard profile: `Mikrotik-Rate-Limit = 50M/50M` `Framed-Pool = pool-active`	10.100.0.0/19 (Normal Pool)	Full Internet: Standard traffic flow. Unlimited routing through default WAN uplink.

EXPIRED	Returns redirection policy: `Framed-Pool = pool-expired` or `Mikrotik-Address-List = EXPIRED`	10.100.32.0/20 (Walled Garden Pool)	Firewall Restricted: Port 80/443 redirected to Nepal Airlink Portal . Customer sees renewal invoice.
RECHARGED	Airlink updates MariaDB (`radreply`) Sends RADIUS CoA / Disconnect packet to NAS	Re-authenticates into: 10.100.0.0/19	Zero-Touch Reconnection: Router reconnects within 2 seconds. Full Internet automatically restored.

4. Technical Implementation Blueprint on MikroTik & FreeRADIUS

To achieve this exact workflow without putting IP pool management strain on FreeRADIUS, use either of the two standard methods below:

Method A: Dual IP Pools via `Framed-Pool` (Direct Match to Diagram)

1. Configure Pools and Firewall on MikroTik:

```
/ip pool add name="pool-active" ranges=10.100.0.2-10.100.31.254
/ip pool add name="pool-expired" ranges=10.100.32.2-10.100.47.254

# Redirect HTTP to Nepal Airlink Portal for expired pool:
/ip firewall nat add chain=dstnat src-address=10.100.32.0/20 protocol=tcp dst-port=80 \
action=dst-nat to-addresses=PORTAL_SERVER_IP to-ports=80
# Block non-portal traffic for expired pool:
/ip firewall filter add chain=forward src-address=10.100.32.0/20 dst-address=!PORTAL_SERVER_IP
action=drop
```

2. FreeRADIUS Response (`radreply` SQL table):

- For Active Subscribers: `INSERT INTO radreply (username, attribute, op, value) VALUES ('user', 'Framed-Pool', '=', 'pool-active');`
- For Expired Subscribers: `INSERT INTO radreply (username, attribute, op, value) VALUES ('user', 'Framed-Pool', '=', 'pool-expired');`

Method B: Single Pool via `Mikrotik-Address-List` (Recommended Best Practice ■)

MikroTik assigns an IP from one single unified pool. If the subscriber is expired, FreeRADIUS sends the vendor attribute **`Mikrotik-Address-List = EXPIRED`**. MikroTik automatically inserts the customer's IP into firewall address-list **`EXPIRED`**. The router firewall redirects the **`EXPIRED`** list to your portal.

```
# Single NAT redirect rule matching the dynamic RADIUS address-list:
/ip firewall nat add chain=dstnat src-address-list=EXPIRED protocol=tcp dst-port=80,443 \
action=dst-nat to-addresses=PORTAL_SERVER_IP to-ports=80
/ip firewall filter add chain=forward src-address-list=EXPIRED dst-address=!PORTAL_SERVER_IP
action=drop
```

5. Summary & Executive Takeaways

1. **IP Ownership:** The NAS (MikroTik) holds physical IP pools (`10.100.0.0/19` and `10.100.32.0/20`). It leases IPs instantly and recovers them on disconnect without DB sync risks.
2. **Policy Ownership:** FreeRADIUS decides *which* pool or address-list the subscriber receives by evaluating their expiration timestamp.
3. **Automated Recovery (CoA):** Upon successful payment in the portal, Airlink triggers a RADIUS Disconnect-Request (RFC 3576). The customer router reconnects in under 2 seconds with the active profile.